

Entropy and Mutual Information

Lecture 2 — Information-theoretic tools for neural coding

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Abstract

We develop Shannon's information theory from the ground up: surprise and entropy, the asymptotic equipartition property and the source-coding bound, the convexity inequalities (Jensen, log-sum, Gibbs), joint and conditional entropy, relative entropy, and mutual information with all its identities. We then go beyond pairwise information to the *multivariate* theory — interaction information, total correlation, and the partial information decomposition of a multi-source channel into *redundant*, *unique*, and *synergistic* components. We treat continuous variables (differential entropy, maximum-entropy distributions, the Gaussian channel and water-filling), the data-processing and Fano inequalities, and close with a detailed application to neural coding: the noise/response decomposition, the direct method and its finite-sample bias, the Fisher-information lower bound, and synergy and redundancy in neural populations.

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1 Surprise, entropy, and compression

Let X be a discrete random variable on an alphabet $\mathcal{X}$ with pmf $p(x)$. The *surprise* of an outcome is $h(x) = -\log p(x)$: it is additive over independent events and decreasing in probability, the only such functional up to a base. With base 2 the unit is the *bit*.

Definition 1.1 (Entropy). $H(X) = \mathbb{E}[-\log p(X)] = -\sum_x p(x) \log p(x)$, with $0 \log 0 = 0$.

Proposition 1.1 (Basic properties). (i) $H(X) \geq 0$, with equality iff X is deterministic; (ii) $H(X) \leq \log |\mathcal{X}|$, with equality iff X is uniform; (iii) H is concave in p .

Example 1.1 (Binary entropy). $X \sim \text{Bernoulli}(p)$ has $H_2(p) = -p \log p - (1-p) \log(1-p)$, maximal (1 bit) at $p = \frac{1}{2}$.

Why entropy is *the* measure: source coding. Entropy is not an arbitrary functional; it is the fundamental limit of lossless compression. By the **asymptotic equipartition property** (AEP), for i.i.d. $X_1, \dots, X_n$,

$$-\frac{1}{n} \log p(X_1, \dots, X_n) \xrightarrow{\text{a.s.}} H(X), \quad (1)$$

so the probability mass concentrates on a *typical set* of $\approx 2^{nH(X)}$ roughly equiprobable sequences. Shannon's source-coding theorem [2, 3] then states that the expected code length L of any uniquely decodable binary code obeys

$$H(X) \leq L < H(X) + 1 \text{ bits/symbol}, \quad (2)$$

the lower bound following from the **Kraft inequality** $\sum_x 2^{-\ell(x)} \leq 1$ together with Gibbs' inequality. Entropy is therefore the average number of bits needed to describe X .

2 Convexity tools

Lemma 2.1 (Jensen). For convex φ , $\mathbb{E}[\varphi(X)] \geq \varphi(\mathbb{E}[X])$ (reversed for concave φ); equality iff X is a.s. constant or φ is affine on the support.

Lemma 2.2 (Log-sum inequality). For nonnegative a_i, b_i , $\sum_i a_i \log \frac{a_i}{b_i} \geq \left(\sum_i a_i\right) \log \frac{\sum_i a_i}{\sum_i b_i}$, with equality iff a_i/b_i is constant.

These two facts generate essentially every inequality below.

3 Joint and conditional entropy

For $(X, Y) \sim p(x, y)$ define $H(X, Y) = -\sum_{x,y} p(x, y) \log p(x, y)$ and the conditional entropy

$$H(Y | X) = \sum_x p(x) H(Y | X=x) = -\sum_{x,y} p(x, y) \log p(y | x). \quad (3)$$

Theorem 3.1 (Chain rule). $H(X, Y) = H(X) + H(Y | X) = H(Y) + H(X | Y)$, and more generally $H(X_1, \dots, X_n) = \sum_i H(X_i | X_1, \dots, X_{i-1})$.

Corollary 3.2 (Conditioning reduces entropy; subadditivity). $H(Y | X) \leq H(Y)$ and hence $H(X_1, \dots, X_n) \leq \sum_i H(X_i)$, with equality iff independent.

4 Relative entropy (Kullback–Leibler divergence)

Definition 4.1. For $p \ll q$, $D_{\text{KL}}(p \parallel q) = \sum_x p(x) \log \frac{p(x)}{q(x)} = \mathbb{E}_p \left[\log \frac{p(X)}{q(X)} \right]$.

Theorem 4.1 (Gibbs' inequality). $D_{\text{KL}}(p \parallel q) \geq 0$, with equality iff $p = q$.

Proof. By Jensen with the concave $\log$, $-D_{\text{KL}}(p \parallel q) = \mathbb{E}_p[\log \frac{q}{p}] \leq \log \mathbb{E}_p[\frac{q}{p}] = \log \sum_x q(x) \leq 0$; strict concavity forces $p = q$ at equality. $\square$

D_{KL} is the inefficiency (extra bits) of coding for q when the truth is p ; it is asymmetric and not a metric, but it dominates the total-variation distance via Pinsker's inequality $D_{\text{KL}}(p \parallel q) \geq \frac{1}{2 \ln 2} \|p - q\|_1^2$.

5 Mutual information

Definition 5.1. $I(X; Y) = D_{\text{KL}}(p(x, y) \parallel p(x)p(y)) = \sum_{x, y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}$.

Derivation of the identities. Start from the definition and split the logarithm using $p(x, y) = p(x | y)p(y)$, so $\log \frac{p(x, y)}{p(x)p(y)} = \log p(x | y) - \log p(x)$:

$$\begin{aligned} I(X; Y) &= \sum_{x, y} p(x, y) \left[\log p(x | y) - \log p(x) \right] \\ &= - \underbrace{\sum_x \left(\sum_y p(x, y) \right) \log p(x)}_{= H(X)} - \underbrace{\left(- \sum_{x, y} p(x, y) \log p(x | y) \right)}_{= H(X|Y)} = H(X) - H(X | Y), \end{aligned} \quad (4)$$

where the first sum used $\sum_y p(x, y) = p(x)$. By the symmetry $p(x | y)p(y) = p(y | x)p(x)$ the same steps give $H(Y) - H(Y | X)$, and substituting the chain rule $H(X | Y) = H(X, Y) - H(Y)$ gives the third form. Hence

$$\boxed{I(X; Y) = H(X) - H(X | Y) = H(Y) - H(Y | X) = H(X) + H(Y) - H(X, Y).} \quad (5)$$

Theorem 5.1 (Properties). (i) $I(X; Y) = I(Y; X) \geq 0$, $= 0$ iff $X \perp Y$; (ii) $I(X; Y) \leq \min\{H(X), H(Y)\}$; (iii) $I(X; X) = H(X)$.

6 Beyond pairs: conditional and interaction information

The *conditional mutual information* is $I(X; Y | Z) = H(X | Z) - H(X | Y, Z) \geq 0$, and mutual information obeys its own chain rule $I(X; Y_1, Y_2) = I(X; Y_1) + I(X; Y_2 | Y_1)$.

Definition 6.1 (Interaction information / co-information). For three variables,

$$\Pi(X; Y; Z) = I(X; Y) - I(X; Y | Z) = I(X; Z) - I(X; Z | Y). \quad (6)$$

Unlike pairwise I , the interaction information **can be negative**. Conditioning can either *decrease* the shared information ($\Pi > 0$, net **redundancy**: Z explains away part of the X – Y dependence) or *increase* it ($\Pi < 0$, net **synergy**: X and Y together reveal more about Z than separately, the classic example being $Z = X \oplus Y$ with independent bits X, Y). The multivariate generalisation is the **total correlation** $C(X_1, \dots, X_n) = \sum_i H(X_i) - H(X_1, \dots, X_n) \geq 0$.

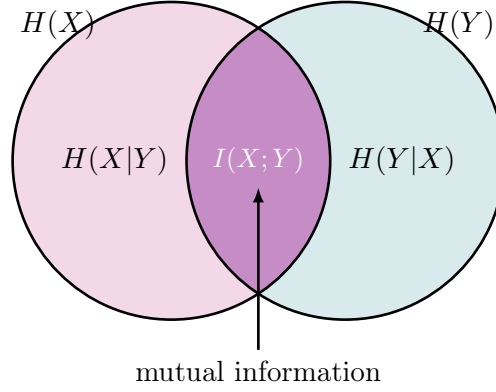


Figure 1: Information diagram. The shaded lens where the two circles overlap is the mutual information $I(X; Y)$; the crescents are the conditional entropies $H(X | Y)$ and $H(Y | X)$, and the total shaded area is the joint entropy $H(X, Y)$.

7 Synergy and redundancy: partial information decomposition

Interaction information conflates synergy and redundancy into a single (signed) number. The **partial information decomposition** (PID; Williams & Beer, 2010) separates them. Consider two sources X_1, X_2 informing about a target Y [9]. The PID posits four non-negative atoms:

$$I(Y; X_1, X_2) = \underbrace{R}_{\text{redundant}} + \underbrace{U_1}_{\text{unique to } X_1} + \underbrace{U_2}_{\text{unique to } X_2} + \underbrace{S}_{\text{synergistic}} \quad (7)$$

constrained to be consistent with the single-source informations,

$$I(Y; X_1) = R + U_1, \quad I(Y; X_2) = R + U_2. \quad (8)$$

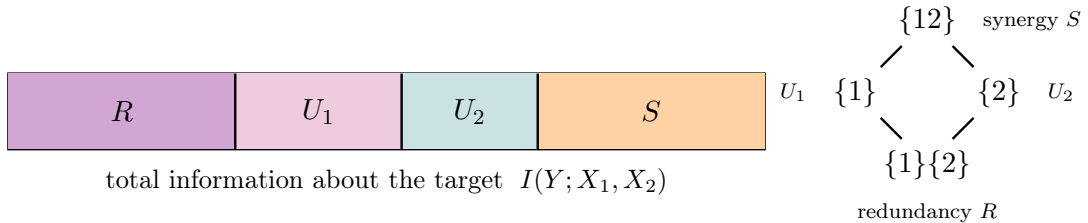


Figure 2: Left: the four PID atoms partition the total information $I(Y; X_1, X_2)$. Right: the redundancy lattice of source collections for two inputs; the four atoms sit on its nodes (bottom = redundancy, middle = unique, top = synergy).

The three identities (7)–(8) are *four* unknowns in three equations: one more definition is required, and there is **no universally agreed choice**. Defining a redundancy measure R fixes the rest via

$$U_i = I(Y; X_i) - R, \quad S = I(Y; X_1, X_2) - I(Y; X_1) - I(Y; X_2) + R = \Pi + R, \quad (9)$$

which makes precise the slogan *synergy minus redundancy = co-information*, $S - R = -\Pi(Y; X_1, X_2)$. Several axiomatic redundancy measures exist — the original $I_{\min}$ of Williams & Beer, the broja measure $I_{\cap}^{\text{BROJA}}$ from an optimisation over consistent joint distributions, and I_{ccs} (common change in surprisal) — agreeing on the two-source case under the Williams–Beer axioms (symmetry, self-redundancy, monotonicity) but differing for $n \geq 3$. PID is an active research front, and is increasingly used in neuroscience to ask whether two brain regions, neurons, or frequency bands carry *redundant*, *unique*, or *synergistic* information about a stimulus or behaviour.

8 Continuous variables

For a density $f(x)$ the *differential entropy* is $h(X) = -\int f \log f \, dx$ (which may be negative and is not coordinate-free), while mutual information $I(X; Y) = h(X) - h(X | Y)$ remains well defined and invariant under smooth invertible maps.

Differential entropy of a Gaussian, derived. For $X \sim \mathcal{N}(0, \sigma^2)$, $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/2\sigma^2}$ so $-\ln f(x) = \frac{1}{2} \ln(2\pi\sigma^2) + \frac{x^2}{2\sigma^2}$. Taking the expectation,

$$h(X) = \mathbb{E}[-\ln f(X)] = \frac{1}{2} \ln(2\pi\sigma^2) + \frac{\mathbb{E}[X^2]}{2\sigma^2} = \frac{1}{2} \ln(2\pi\sigma^2) + \frac{1}{2} = \frac{1}{2} \ln(2\pi e \sigma^2), \quad (10)$$

using $\mathbb{E}[X^2] = \sigma^2$ (and $\log_2$ replaces $\ln$ to convert nats to bits).

Maximum-entropy distributions, derived. Maximise $h[f] = -\int f \ln f \, dx$ subject to $\int f \, dx = 1$ and $\int x^2 f \, dx = \sigma^2$. Form the Lagrangian with multipliers λ_0, λ_2 and take the functional derivative with respect to $f(x)$:

$$\frac{\delta}{\delta f} \left[-\int f \ln f + \lambda_0 \int f + \lambda_2 \int x^2 f \right] = -\ln f(x) - 1 + \lambda_0 + \lambda_2 x^2 \stackrel{!}{=} 0, \quad (11)$$

so $f^*(x) = \exp(\lambda_0 - 1 + \lambda_2 x^2) \propto e^{\lambda_2 x^2}$: a Gaussian (with $\lambda_2 < 0$). Fixing the two multipliers by the constraints recovers $\mathcal{N}(0, \sigma^2)$. Hence **among all densities of variance σ^2 the Gaussian maximises differential entropy**, $h \leq \frac{1}{2} \log(2\pi e \sigma^2)$. The same Lagrange recipe gives the exponential family $f^* \propto \exp(\sum_k \lambda_k T_k)$ for moment constraints $\mathbb{E}[T_k]$ (e.g. the exponential distribution on $[0, \infty)$ for fixed mean, the uniform on a bounded interval).

The Gaussian channel, derived. Let $Y = X + N$ with $N \sim \mathcal{N}(0, \sigma^2)$ independent of X and power constraint $\mathbb{E}[X^2] \leq P$. *Step 1:* since $Y | X$ is just N shifted by the constant X , and differential entropy is shift-invariant, $h(Y | X) = h(N) = \frac{1}{2} \log(2\pi e \sigma^2)$. *Step 2:* $I(X; Y) = h(Y) - h(Y | X) = h(Y) - \frac{1}{2} \log(2\pi e \sigma^2)$, so we maximise $h(Y)$. *Step 3:* $\text{Var}(Y) = \mathbb{E}[X^2] + \sigma^2 \leq P + \sigma^2$, and by the maximum-entropy result $h(Y)$ is largest when Y is Gaussian, i.e. when X is Gaussian, giving $h(Y) \leq \frac{1}{2} \log(2\pi e (P + \sigma^2))$. *Step 4:* subtract:

$$C = \max_{\mathbb{E}[X^2] \leq P} I(X; Y) = \frac{1}{2} \log(2\pi e (P + \sigma^2)) - \frac{1}{2} \log(2\pi e \sigma^2) = \boxed{\frac{1}{2} \log\left(1 + \frac{P}{\sigma^2}\right)}. \quad (12)$$

For n parallel independent Gaussian channels with noise powers σ_i^2 , maximising $\sum_i \frac{1}{2} \log(1 + P_i/\sigma_i^2)$ under $\sum_i P_i = P$ with a Lagrange multiplier ν gives the **water-filling** solution $P_i = (\nu - \sigma_i^2)^+$: power is poured into the quietest channels first.

9 Data processing, sufficiency, and Fano

Theorem 9.1 (Data-processing inequality). *If $X \rightarrow Y \rightarrow Z$ is Markov ($Z \perp X | Y$) then $I(X; Z) \leq I(X; Y)$.*

Proof. $I(X; Y, Z) = I(X; Y) + I(X; Z | Y) = I(X; Z) + I(X; Y | Z)$; the Markov property kills $I(X; Z | Y)$ and $I(X; Y | Z) \geq 0$. $\square$

Equality holds iff Y is a *sufficient statistic* for X . No processing of Y creates information about X : a downstream neuron cannot recover more about a stimulus than its afferents deliver.

Theorem 9.2 (Fano's inequality). *For any estimator $\hat{X} = g(Y)$ of X with error probability $P_e = \Pr(\hat{X} \neq X)$,*

$$H(X | Y) \leq H_2(P_e) + P_e \log(|\mathcal{X}| - 1). \quad (13)$$

Fano converts a bound on residual entropy (hence on $I(X; Y)$) into a bound on decoding error, linking information to achievable read-out accuracy.

10 Application: information in neural responses

Let $S \sim p(s)$ be a stimulus and R the neural response (spike count, binned spike train, or continuous feature). The coding quantity of interest is

$$I(S; R) = \underbrace{H(R)}_{\text{response entropy}} - \underbrace{\mathbb{E}_S[H(R | S)]}_{\text{noise entropy}} : \quad (14)$$

a neuron is informative when responses are *diverse* across stimuli yet *reliable* for each stimulus.

Direct method and bias. Binning time at resolution Δt over L bins yields binary words $w \in \{0, 1\}^L$. Estimating the total and noise entropy rates from many stimulus repetitions gives the Strong-de Ruyter-Bialek estimate $I_{\text{rate}} = H_{\text{tot}} - H_{\text{noise}}$ (bits/s) [6, 5]. Plug-in entropy estimates are biased; the leading (Miller-Madow / Panzeri-Treves) bias is

$$\langle \hat{I} \rangle - I \approx -\frac{1}{2N \ln 2} \left[\sum_s (\tilde{R}_s - 1) - (\tilde{R} - 1) \right], \quad (15)$$

where $\tilde{R}_s$ is the number of response bins occupied given stimulus s and $\tilde{R}$ the number occupied overall: the bias is of order $(\# \text{states})/N$ for N trials, motivating bias correction (Panzeri-Treves, NSB estimator) [7] or quadratic extrapolation in $1/N$.

Fisher-information lower bound. For a continuous parameter θ (e.g. stimulus orientation) encoded by response r with likelihood $p(r | \theta)$, the **Fisher information** $J(\theta) = \mathbb{E}[(\partial_\theta \log p(r | \theta))^2]$ bounds any unbiased estimator through Cramér-Rao, $\text{Var}(\hat{\theta}) \geq 1/J(\theta)$. In the Gaussian/large- N regime the mutual information is captured by

$$I(\theta; r) \gtrsim H(\theta) - \frac{1}{2} \mathbb{E}_\theta \left[\log \frac{2\pi e}{J(\theta)} \right], \quad (16)$$

the Brunel-Nadal bound [8], tying coding efficiency to the sharpness of tuning curves ($J \propto (f'(\theta))^2 / \text{noise}$).

Populations: synergy and redundancy. For a population $\mathbf{r} = (r_1, \dots, r_n)$ the PID and interaction information of Sections 6–7 ask whether neurons are *redundant* (overlapping codes, robust but inefficient), carry *unique* information, or are *synergistic* (the joint pattern means more than the parts). **Noise correlations** — shared trial-to-trial variability — can raise or lower $I(S; \mathbf{r})$ relative to independent neurons depending on their sign relative to the signal correlations, the central issue in the population-coding literature.

Efficient coding. Maximising $I(S; R)$ over the response distribution under metabolic and noise constraints predicts that optimal neurons should *decorrelate* and *equalise* their outputs (histogram equalisation, Laughlin 1981) and match their tuning to the stimulus statistics — the information-theoretic foundation of the efficient-coding hypothesis [10, 11].

11 Sequences and the asymptotic equipartition property

The deep content of information theory appears when we encode *long sequences*. The following results (the AEP) formalise the idea that, although an alphabet has $|\mathcal{X}|^n$ possible sequences, almost all probability concentrates on a much smaller *typical set* of $\approx 2^{nH}$ sequences [1, 3].

Theorem 11.1 (Weak AEP). *Let $X_1, \dots, X_n$ be i.i.d. copies of X . Then $-\frac{1}{n} \log p(X_1, \dots, X_n) \xrightarrow{\text{prob}} H(X)$ as $n \rightarrow \infty$.*

Proof. By independence, $-\frac{1}{n} \log p(X_1, \dots, X_n) = \frac{1}{n} \sum_{i=1}^n (-\log p(X_i))$, an average of i.i.d. terms with mean $\mathbb{E}[-\log p(X)] = H(X)$. The weak law of large numbers gives convergence in probability to $H(X)$. $\square$

Definition 11.1 (Typical set). For $\epsilon > 0$, $T_\epsilon^{(n)} = \{x^n \in \mathcal{X}^n : 2^{-n(H+\epsilon)} \leq p(x^n) \leq 2^{-n(H-\epsilon)}\}$.

Theorem 11.2 (Properties of the typical set). *For every $\epsilon > 0$ and n large enough: (i) $\Pr(X^n \in T_\epsilon^{(n)}) > 1 - \epsilon$; (ii) $|T_\epsilon^{(n)}| \leq 2^{n(H+\epsilon)}$; (iii) $|T_\epsilon^{(n)}| \geq (1 - \epsilon)2^{n(H-\epsilon)}$.*

Proof. (i) is the Weak AEP applied to the event defining $T_\epsilon^{(n)}$. (ii): since every $x^n \in T_\epsilon^{(n)}$ has $p(x^n) \geq 2^{-n(H+\epsilon)}$, $1 \geq \sum_{x^n \in T_\epsilon^{(n)}} p(x^n) \geq |T_\epsilon^{(n)}| 2^{-n(H+\epsilon)}$, giving the bound. (iii): by (i), $1 - \epsilon < \Pr(T_\epsilon^{(n)}) \leq |T_\epsilon^{(n)}| 2^{-n(H-\epsilon)}$. $\square$

Thus $\approx 2^{nH}$ sequences carry essentially all the probability and can be indexed with $\approx nH$ bits — the intuition behind the source-coding theorem below.

12 Source coding I: the Kraft–McMillan theorem

A symbol code $c : \mathcal{X} \rightarrow \mathcal{Y}^*$ maps each symbol to a finite string over an alphabet of size $d = |\mathcal{Y}|$; it is *uniquely decodable* if distinct messages have distinct encodings, and *prefix* (instantaneous) if no codeword is a prefix of another.

Theorem 12.1 (Kraft–McMillan). *(1) Any uniquely decodable code with lengths $\ell_x = |c(x)|$ satisfies $\sum_x d^{-\ell_x} \leq 1$. (2) Conversely, given lengths (ℓ_x) with $\sum_x d^{-\ell_x} \leq 1$, there exists a prefix code with those lengths.*

Proof. (1) For any n , expand the n -th power $(\sum_x d^{-\ell_x})^n = \sum_{x_1, \dots, x_n} d^{-(\ell_{x_1} + \dots + \ell_{x_n})} = \sum_{k=1}^{n\ell_{\max}} a(k) d^{-k}$, where $a(k)$ counts messages of total encoded length k . Unique decodability means distinct messages give distinct strings, so $a(k) \leq d^k$ (there are only d^k strings of length k); hence $(\sum_x d^{-\ell_x})^n \leq n\ell_{\max} d^{n\ell_{\max}}$ and, taking n -th roots and $n \rightarrow \infty$, $\sum_x d^{-\ell_x} \leq 1$. (2) Order $\ell_1 \leq \dots \leq \ell_m$ and set $r_m = \sum_{i \leq m} d^{-\ell_i} \leq 1$; let $c(m)$ be the first ℓ_m digits of the d -ary expansion of r_m . If $c(m)$ were a prefix of $c(n)$ ($m < n$) then $r_n - r_m < d^{-\ell_m}$, yet by construction $r_n - r_m = \sum_{i=m+1}^n d^{-\ell_i} \geq d^{-\ell_m}$ — a contradiction. So the code is prefix. $\square$

Corollary 12.2. *For any uniquely decodable code there is a prefix code with the same lengths: restricting to prefix codes costs nothing.*

13 Source coding II: Shannon’s first theorem and optimal codes

Theorem 13.1 (Source-coding bound for symbol codes). *For a uniquely decodable d -ary code, $H_d(X) \leq \mathbb{E}|c(X)|$, with equality iff $\ell_x = -\log_d p(x)$.*

Proof. Let $\ell_x = |c(x)|$ and $q(x) = d^{-\ell_x}/Z$ with $Z = \sum_{x'} d^{-\ell_{x'}} \leq 1$ (Kraft). Then

$$\begin{aligned} \mathbb{E}|c(X)| - H_d(X) &= \sum_x p(x)\ell_x + \sum_x p(x) \log_d p(x) = \sum_x p(x) \log_d \frac{p(x)}{d^{-\ell_x}} \\ &= \sum_x p(x) \log_d \frac{p(x)}{q(x)} - \log_d Z = D_{\text{KL}}(p \| q) - \log_d Z \geq 0, \end{aligned} \quad (17)$$

since $D_{\text{KL}} \geq 0$ (Theorem 4.1) and $Z \leq 1$. Equality forces $p = q$ and $Z = 1$, i.e. $\ell_x = -\log_d p(x)$. $\square$

The optimum $\ell_x = -\log_d p(x)$ is generally non-integer. Rounding up ($\ell_x = \lceil -\log_d p(x) \rceil$, the **Shannon code**) gives $H_d \leq \mathbb{E}[c] < H_d + 1$, an overhead of up to one symbol. **Block coding** removes it: encoding n symbols at once yields $H_d(X) \leq \frac{1}{n} \mathbb{E}[c(X^n)] < H_d(X) + \frac{1}{n} \xrightarrow{n \rightarrow \infty} H_d(X)$ — *Shannon's first (source-coding) theorem*. Several constructions are standard [1, 4]:

- **Huffman (1952)** — provably *optimal* prefix code; build the code tree by repeatedly merging the two least probable symbols. *Example*: $p = (0.4, 0.2, 0.2, 0.1, 0.1)$, $H \approx 2.12$ bits, Huffman lengths $(2, 2, 2, 3, 3)$, $L = 2.2$ bits (within 4% of H).
- **Shannon (1948)** and **Fano (1949)** — near-optimal length assignments from the cumulative distribution.
- **Arithmetic / Elias coding** — encode the *whole message* as a sub-interval of $[0, 1)$; approaches H without the per-symbol rounding loss and handles adaptive models.
- **Tunstall (1967)** — a variable-to-fixed dual that maps variable-length source blocks to fixed-length codewords.

14 Channel coding: Shannon's second theorem

A **discrete memoryless channel** (DMC) is a conditional law $p(y | x)$ applied independently to each symbol. Its **capacity** is

$$C = \max_{p(x)} I(X; Y) \quad [\text{bits per channel use}]. \quad (18)$$

Examples. Binary symmetric channel with crossover f : $C = 1 - H_2(f)$ (so $C = 1$ at $f = 0$, $C = 0$ at $f = \frac{1}{2}$). Binary erasure channel with erasure probability e : $C = 1 - e$.

An (m, n) code maps m messages to length- n inputs and decodes outputs back to messages; its **rate** is $\rho = \frac{\log m}{n}$ bits/use, and a rate R is *achievable* if codes exist with $\rho \rightarrow R$ and maximal error $\rightarrow 0$.

Theorem 14.1 (Shannon's second theorem). *For a DMC of capacity C , a rate $R > 0$ is achievable if and only if $R \leq C$.*

Why C (joint-typicality decoding). Generate a random codebook of i.i.d. inputs and decode y^n to the unique codeword that is *jointly typical* with it. Two facts make this work for large n : of the $\approx 2^{nH(Y)}$ typical outputs, each input is compatible with $\approx 2^{nH(Y|X)}$ of them, so the number of distinguishable inputs — whose output sets do not overlap — is at most

$$\frac{2^{nH(Y)}}{2^{nH(Y|X)}} = 2^{n(H(Y) - H(Y|X))} = 2^{nI(X; Y)}, \quad (19)$$

giving a reliable rate of $I(X; Y) \leq C$ bits/use. Optimizing the input distribution attains C .

Converse (why not more). If $R > C$, Fano's inequality forces $H(\text{message} | Y^n)$ to stay bounded away from 0, so the decoding-error probability cannot vanish. Capacity is thus the sharp threshold between reliable and unreliable communication — the channel analogue of entropy.

15 Error-correcting (block linear) codes

Shannon's theorem is non-constructive; *coding theory* builds explicit codes approaching C . A binary $[n, k]$ **linear code** is a k -dimensional subspace of $\{0, 1\}^n$, specified by a generator matrix G (encoding $x \mapsto xG$) or a parity-check matrix H (codewords satisfy $Hc^\top = 0$). Its *minimum distance* d (smallest Hamming weight of a nonzero codeword) sets its power: it corrects up to

$\lfloor (d-1)/2 \rfloor$ bit errors. The **Hamming** $[7, 4]$ **code** ($d = 3$) encodes 4 data bits into 7 and corrects any single-bit error via its syndrome $Hc^\top$; modern turbo and LDPC codes approach the Shannon limit closely.

16 Rate–distortion: lossy compression

When a distortion D is tolerable, the minimal rate is the **rate–distortion function**

$$R(D) = \min_{p(\hat{x}|x): \mathbb{E}[d(X, \hat{X})] \leq D} I(X; \hat{X}). \quad (20)$$

For a Gaussian source $X \sim \mathcal{N}(0, \sigma^2)$ under squared error, $R(D) = \frac{1}{2} \log \frac{\sigma^2}{D}$ for $0 \leq D \leq \sigma^2$ (and 0 beyond): each halving of distortion costs half a bit per sample. This is the lossy counterpart of source coding and the basis of perceptual compression [3].

17 A worked neural-coding example

A neuron emits a spike count R in a fixed window in response to one of two equally likely stimuli $s \in \{1, 2\}$ with Poisson statistics of means $\lambda_1 = 2$ and $\lambda_2 = 6$. Then $p(r | s) = e^{-\lambda_s} \lambda_s^r / r!$ and $p(r) = \frac{1}{2}[p(r | 1) + p(r | 2)]$, and $I(S; R) = H(R) - \frac{1}{2}[H(\text{Pois}(2)) + H(\text{Pois}(6))] \approx 0.6$ bits. A single count window thus already resolves most of the 1-bit stimulus identity, the more so the larger the separation $|\lambda_1 - \lambda_2|$ relative to the Poisson noise $\sqrt{\lambda}$ — the discrete analogue of the d' /Fisher-information analysis of tuning curves.

Exercises

1. Prove $H(X) \leq \log|\mathcal{X}|$ directly from Jensen applied to $-\log$.
2. Give examples where $I(X; Y | Z)$ is (a) larger and (b) smaller than $I(X; Y)$ (hint: common cause vs. XOR).
3. Compute the capacity of the binary erasure channel with erasure probability e .
4. For independent fair bits X_1, X_2 and $Y = X_1 \oplus X_2$, compute all four PID atoms and verify $S = 1$ bit while $I(Y; X_i) = 0$.
5. Derive the Gaussian rate–distortion function $R(D) = \frac{1}{2} \log(\sigma^2/D)$.
6. Estimate the plug-in entropy bias for N samples on an alphabet of size m , and discuss when it matters for neural data.

18 Summary

Entropy measures uncertainty and sets the compression limit; relative entropy measures model mismatch; mutual information (5) measures shared information; and beyond pairs, interaction information and the partial information decomposition (7) separate *redundant*, *unique*, and *synergistic* contributions of multiple sources. The non-negativity of D_{KL} underlies every inequality; data processing and Fano constrain any read-out. Applied to neurons via (14), with bias correction and the Fisher-information bound, these tools turn “how much does a spike train tell us about the world?” — and “do neurons cooperate or duplicate?” — into measurable quantities, and ground the efficient-coding theory of neural representation.

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